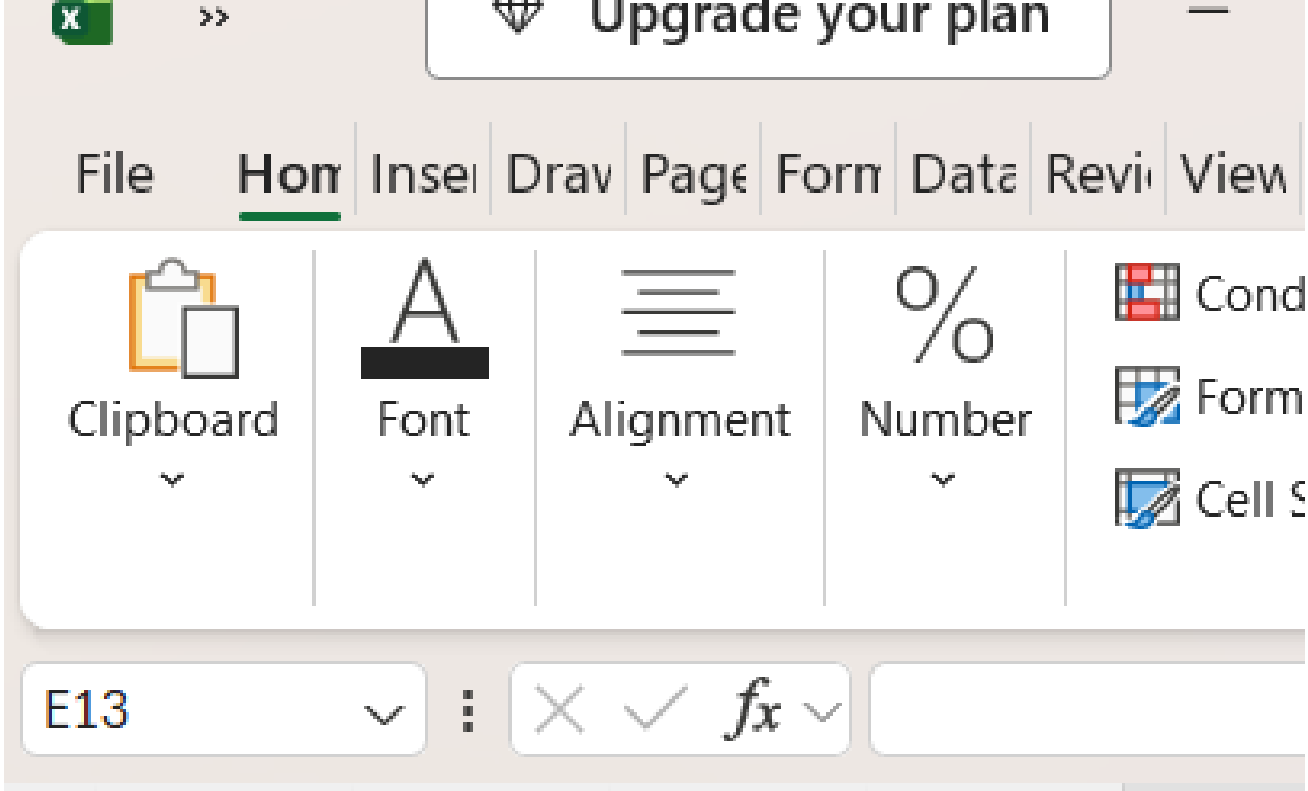
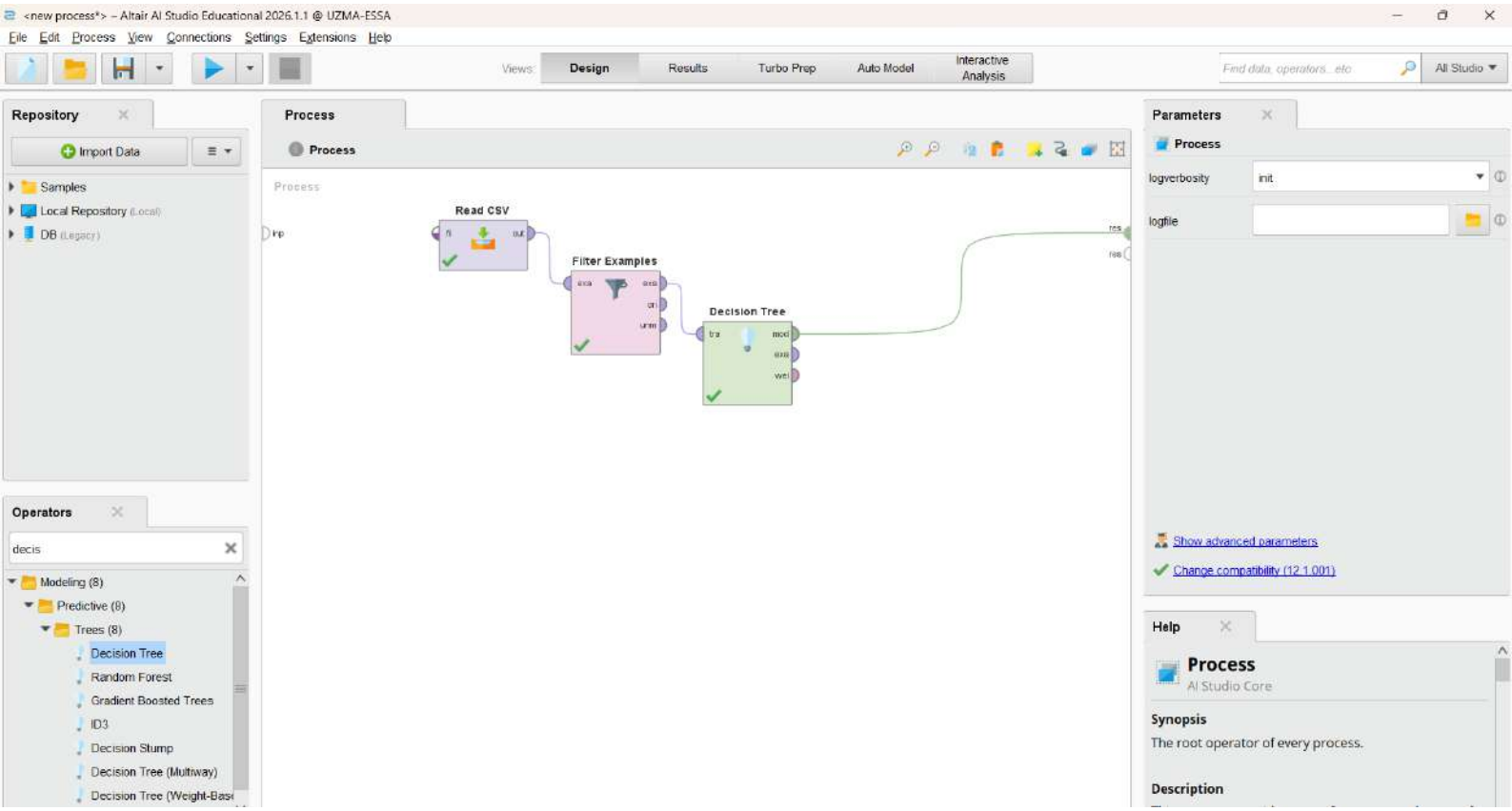
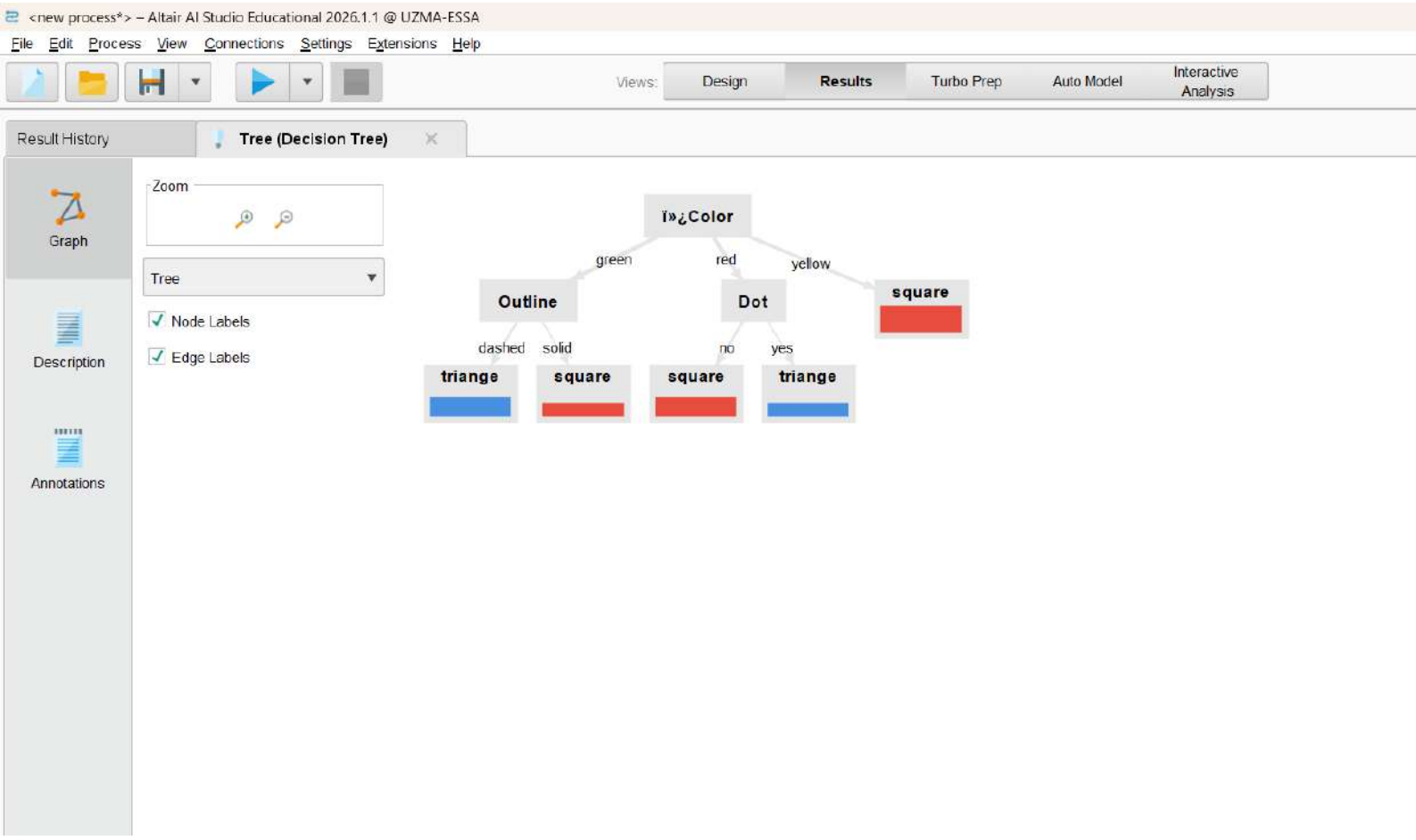


	A	B	C	D	E
1	Color	Outline	Dot	shape	
2	green	dashed	no	triange	
3	green	dashed	yes	triange	
4	yellow	dashed	no	square	
5	red	dashed	no	square	
6	red	solid	no	square	
7	red	solid	yes	triange	
8	green	solid	no	square	
9	green	dashed	no	triange	
10	yellow	solid	yes	square	
11	red	solid	no	square	
12	green	solid	yes	square	
13	yellow	dashed	yes	square	
14	yellow	solid	no	square	
15	red	dashed	yes	triange	
16	green	solid	no		
17	red	dashed	yes		
18					

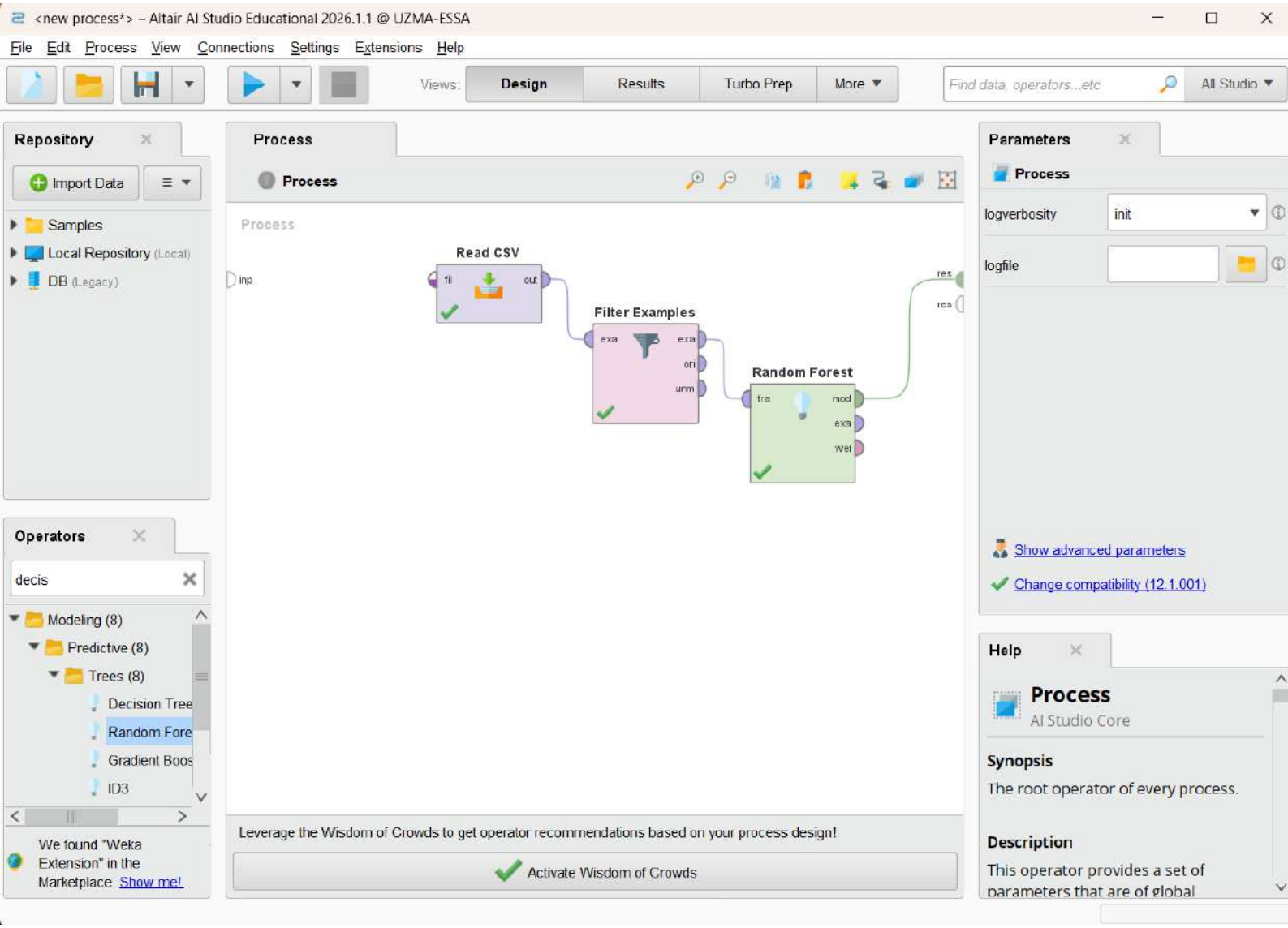
Decision tree



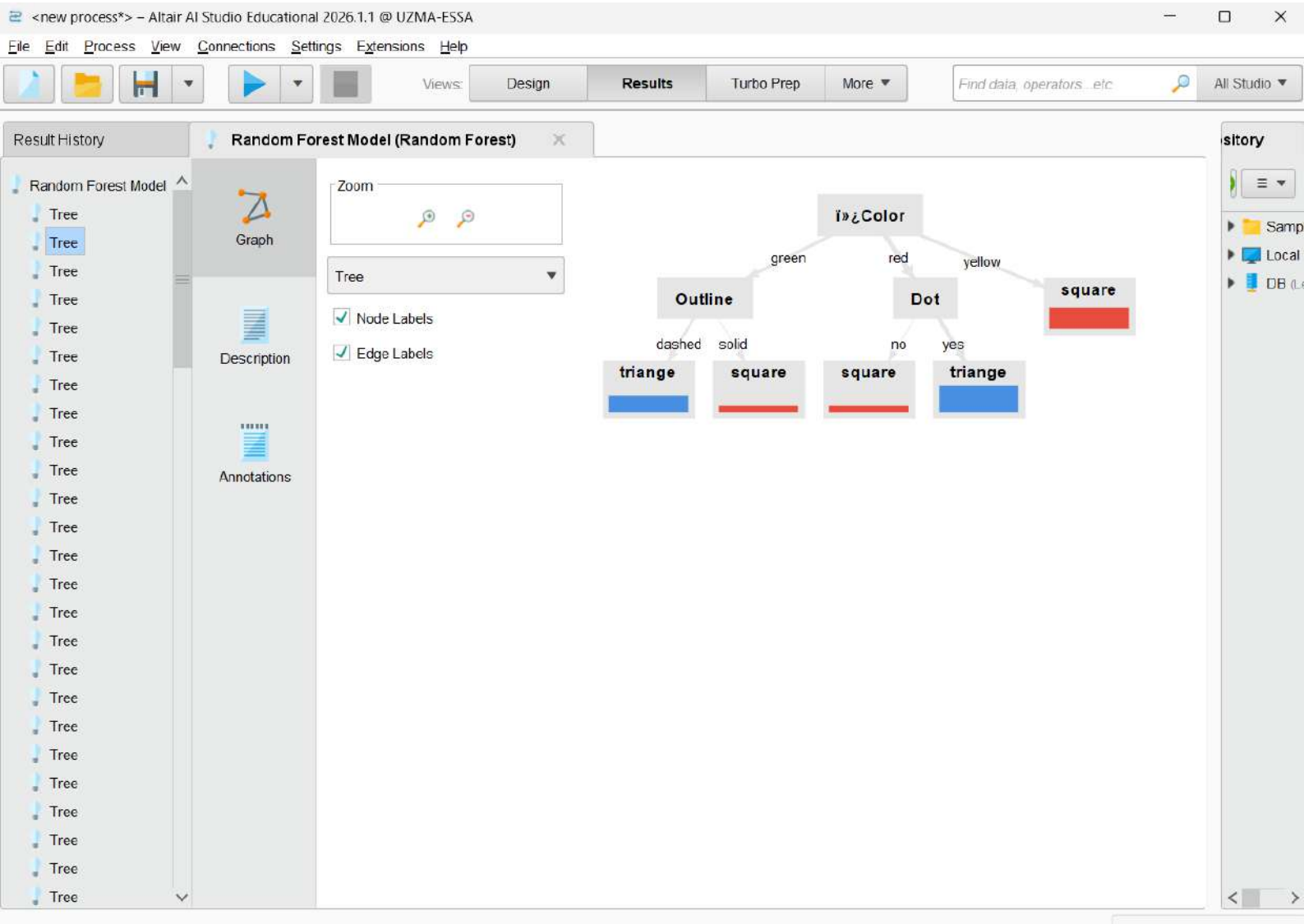
Result Decision Tree



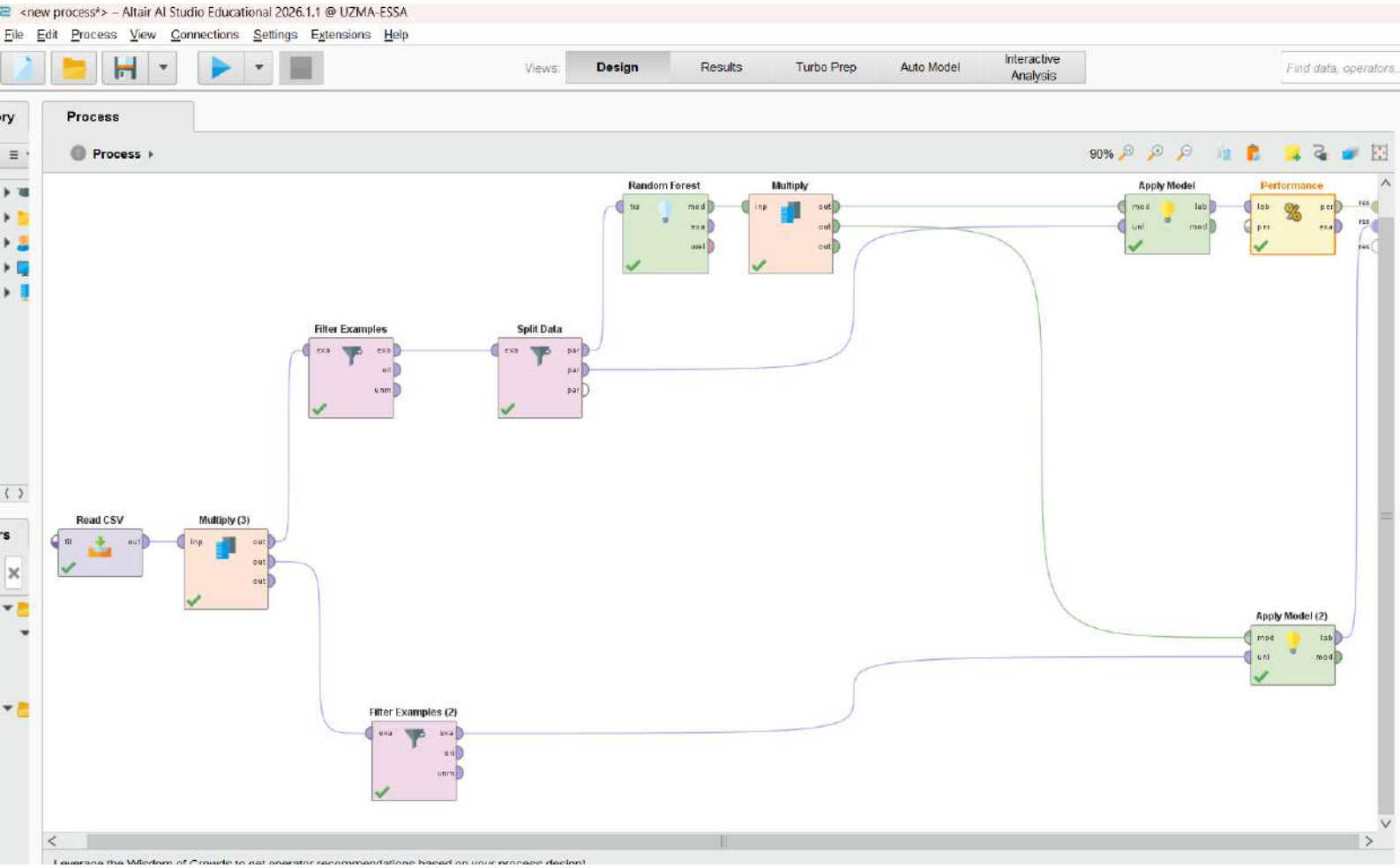
Random forest Design

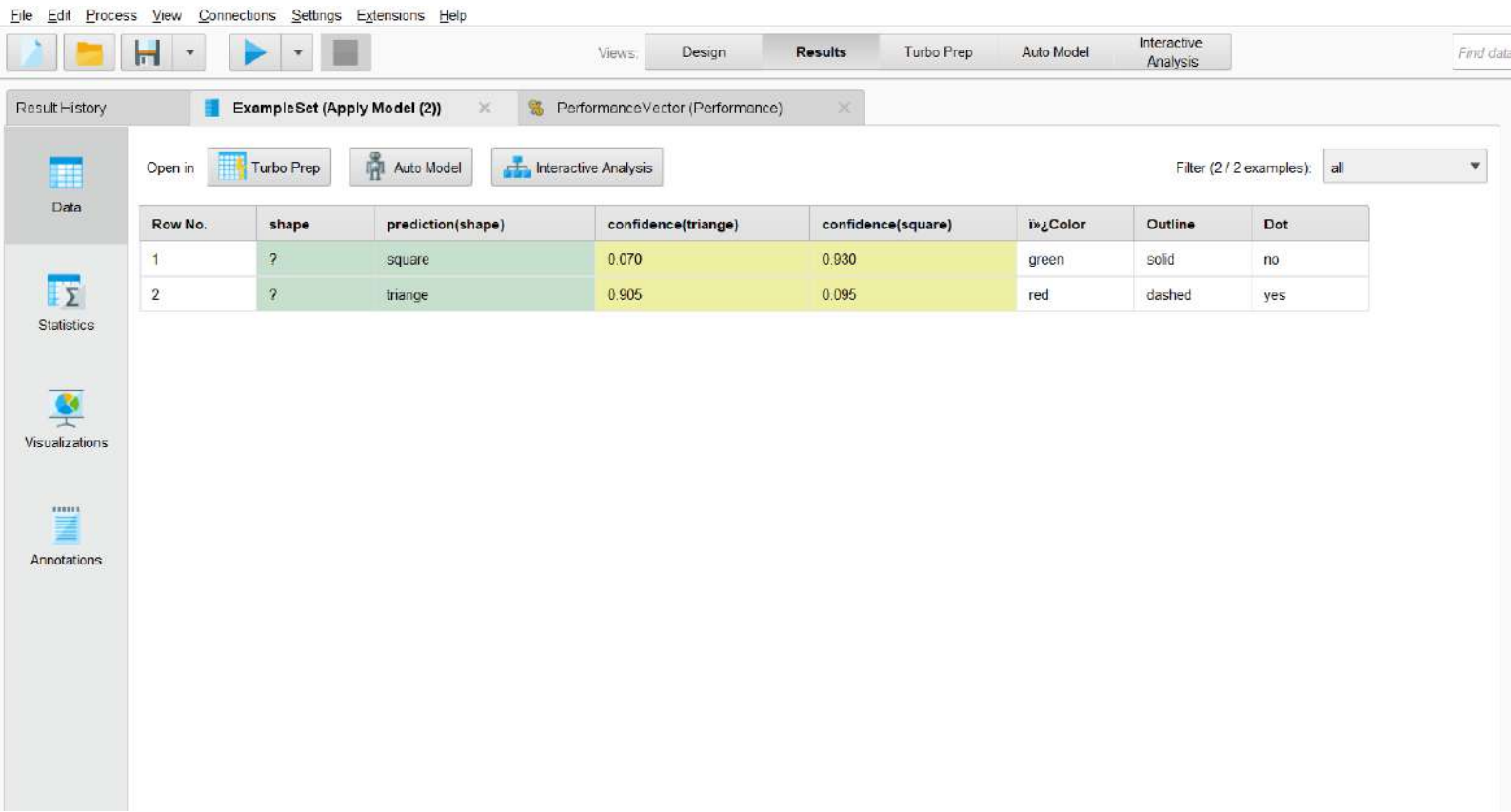


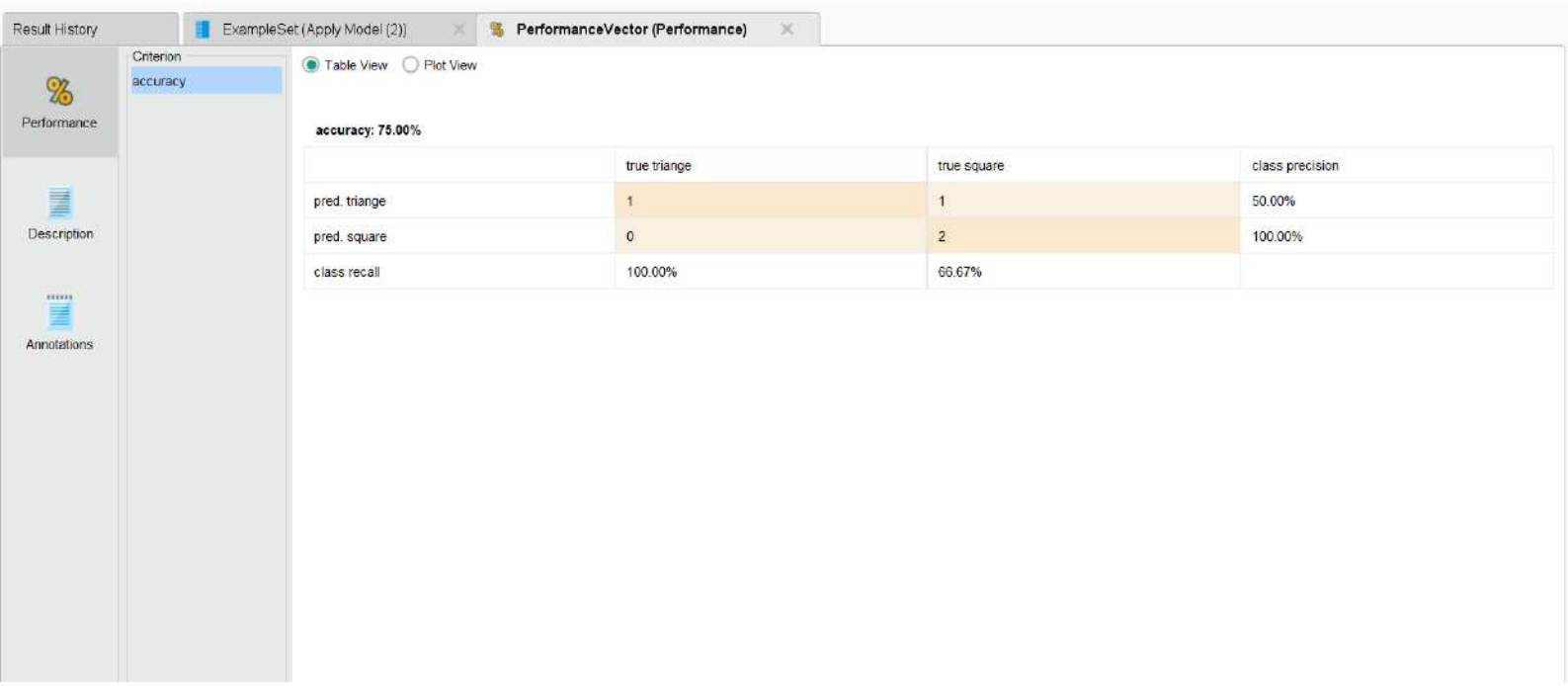
Random Forest Result



Task 2


<new process*> – Altair AI Studio Educational 2026.1.1 @ UZMA-ESSA
File Edit Process View Connections Settings Extensions Help
Views: Design Results Turbo Prep Auto Model Interactive Analysis
Find data, operators...
Process
Process
90%
Random Forest
Multiply
Apply Model
Performance
Filter Examples
Split Data
Read CSV
Multiply (3)
Apply Model (2)
Filter Examples (2)


File Edit Process View Connections Settings Extensions Help
Views: Design Results Turbo Prep Auto Model Interactive Analysis
Result History
ExampleSet (Apply Model (2))
PerformanceVector (Performance)
Data
Statistics
Visualizations
Annotations
Open in Turbo Prep Auto Model Interactive Analysis
Filter (2 / 2 examples): all
Row No. shape prediction(shape) confidence(triange) confidence(square) i»¿Color Outline Dot
1 ? square 0.070 0.930 green solid no
2 ? triange 0.905 0.095 red dashed yes


Result History
ExampleSet (Apply Model (2))
PerformanceVector (Performance)
Performance
Description
Annotations
Criterion
accuracy
Table View
Plot View
accuracy: 75.00%
true triange
true square
class precision
pred. triange
1
1
50.00%
pred. square
0
2
100.00%
class recall
100.00%
66.67%

CONCLUSION

Random Forest was trained on the rows with known shapes and reached 75% accuracy. It predicted the two missing shapes as square and triangle. It predicted Row 1 as square (93%) and Row 2 as triangle (90.5%) because similar rows in the data had those shapes.

Missing shape predictions :

Row 1 (green, solid, no) =square (93% confidence) and triangle(7 % confidence)

Row 2 (red, dashed, yes) =triangle (90.5% confidence) and square (9.5% confidence)